Energy-Aware Token-Level Routing for Heterogeneous LLM Inference in Kubernetes

Design, Implementation, and Evaluation of an `llm-d` Endpoint Picker Plugin

Johnnie Yan Ho Tse

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Abstract

Large Language Model (LLM) inference has rapidly emerged as one of the most significant consumers of electrical energy in modern hyperscale data center operations. Current LLM serving systems and inference schedulers predominantly route requests using heuristics that are optimized for latency reduction or KV-cache reuse. However, these systems fundamentally lack the awareness required to consider the energy cost per generated token or the real-time carbon intensity of the electrical grid powering the compute endpoints. This thesis presents the design, implementation, and evaluation of an energy-aware Endpoint Picker Plugin (EPP) designed for the `llm-d` inference scheduler—a Kubernetes-native framework constructed upon the standard Gateway API Inference Extension (GIE). The proposed plugin introduces a rigorous multi-objective scoring pipeline that simultaneously optimizes for energy efficiency, carbon footprint minimization, and latency Service Level Objective (SLO) compliance. By leveraging an ϵ-constraint method derived from Pareto multi-objective optimization theory, the system isolates latency guarantees as hard constraints while optimizing energy metrics within the remaining feasible solution space.

The system implements five key architectural innovations: (1) a phase-aware energy scorer utilizing distinct weight vectors for prefill and decode inference phases, (2) an SLO constraint filter enforcing Time-To-First-Token (TTFT) and Time-Per-Output-Token (TPOT) bounds, (3) a KV-cache transfer energy model accounting for disaggregated serving network overheads, (4) a Software Carbon Intensity (SCI) calculator aligned with the Green Software Foundation's strict specifications, and (5) an adaptive weight controller that dynamically modulates scoring weights in response to real-time grid carbon signals and cluster power constraints.

Implemented entirely in Go, the plugin is highly optimized, containerized as a minimal 8.6 MB distroless image, and validated within a Kubernetes cluster simulating heterogeneous environments of high-power GPUs and low-power ASICs. Comprehensive evaluation demonstrates that the energy-aware routing policy reduces estimated energy consumption by 17.4% on average and up to 32% for decode-heavy workloads compared to hardware-agnostic round-robin scheduling, while strictly adhering to tail-latency SLOs. Furthermore, the adaptive carbon-aware controller enables dynamic temporal load shifting, demonstrating linear reductions in absolute carbon footprint correlated with regional grid emission factors.

Acknowledgments

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List of Abbreviations

- **API**: Application Programming Interface
- **CapEx**: Capital Expenditure
- **DCGM**: Data Center GPU Manager
- **DVFS**: Dynamic Voltage and Frequency Scaling
- **EDP**: Energy-Delay Product
- **EKS**: Elastic Kubernetes Service (Amazon)
- **EPP**: Endpoint Picker Plugin
- **FSM**: Finite State Machine
- **GIE**: Gateway API Inference Extension
- **GKE**: Google Kubernetes Engine
- **GPU**: Graphics Processing Unit
- **GSF**: Green Software Foundation
- **HBM**: High Bandwidth Memory
- **KEDA**: Kubernetes Event-driven Autoscaling
- **KV-Cache**: Key-Value Cache
- **LLM**: Large Language Model
- **LoC**: Lines of Code
- **MIG**: Multi-Instance GPU
- **NVML**: NVIDIA Management Library
- **OOM**: Out-Of-Memory

- **OpEx**: Operational Expenditure
- **OSDI**: Symposium on Operating Systems Design and Implementation
- **PCIe**: Peripheral Component Interconnect Express
- **RAPL**: Running Average Power Limit (Intel)
- **SCI**: Software Carbon Intensity
- **SLO**: Service Level Objective
- **TCO**: Total Cost of Ownership
- **TDP**: Thermal Design Power
- **TPOT**: Time-Per-Output-Token
- **TTFT**: Time-To-First-Token
- **vLLM**: Virtual Large Language Model (Inference Engine)
- **WVA**: Workload Variant Autoscaler

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Chapter 1

Introduction

The deployment of Large Language Models (LLMs) at hyperscale has triggered a fundamental paradigm shift in cloud computing, reshaping not only software architecture but the physical realities of global datacenter infrastructure. As organizations race to integrate generative artificial intelligence into their core products, the computational locus has aggressively transitioned from model training to model inference. While training a massive transformer model is a singular, highly parallelized event, inference is a continuous, perpetual process serving billions of daily requests. This shift has precipitated an unprecedented energy crisis within the tech industry, challenging the limits of electrical grids, thermal dissipation technologies, and corporate sustainability goals.

This thesis investigates the critical intersection of distributed systems scheduling and hardware thermodynamics. It presents the design, mathematical formulation, and implementation of an energy-aware, phase-aware inference scheduler designed natively for Kubernetes and the Gateway API Inference Extension (GIE).

1.1 The AI Computing Paradigm Shift

The genesis of the current AI boom can be traced to the introduction of the Transformer architecture in 2017, which leveraged self-attention mechanisms to achieve unprecedented contextual understanding in natural language processing. The subsequent years witnessed an exponential scaling of model parameters, colloquially

known as "scaling laws." Models grew from hundreds of millions of parameters (e.g., GPT-1) to hundreds of billions (e.g., GPT-3, Llama-3 70B), and now into the trillions for Mixture-of-Experts (MoE) architectures.

To execute these models, the underlying hardware had to evolve. General-purpose Central Processing Units (CPUs) were quickly rendered obsolete for AI workloads due to their low memory bandwidth and lack of dense matrix multiplication capabilities. The industry pivoted entirely to specialized accelerators, primarily High-Performance Computing (HPC) Graphics Processing Units (GPUs) equipped with Tensor Cores and High Bandwidth Memory (HBM). However, this specialized hardware comes at a staggering physical cost. The transistors required to execute trillions of floating-point operations per second (TFLOPS) draw massive amounts of electrical current, generating immense heat that must be continuously dissipated to prevent catastrophic hardware failure.

1.2 The Environmental and Energy Crisis of LLMs

The operational realities of modern AI datacenters are fundamentally constrained by physics. A single NVIDIA H100 Tensor Core GPU—the current industry standard for state-of-the-art inference—operates with a Thermal Design Power (TDP) of up to 700 Watts. When aggregated into standard server chassis (e.g., an 8-GPU HGX node) and mounted into server racks, the power density exceeds 30 to 40 kilowatts (kW) per rack. Traditional air-cooling infrastructure is incapable of managing this heat density, forcing datacenters to adopt expensive direct-to-chip liquid cooling systems.

At the macro level, the aggregated power draw of hyperscale AI clusters is measured in hundreds of Megawatts (MW), rivaling the electrical consumption of small cities. The International Energy Agency (IEA) has projected that datacenter electricity consumption, driven largely by AI inference, will double between 2022 and 2026. This exponential growth poses a severe threat to global decarbonization efforts. The carbon footprint of an inference request is dictated not only by the absolute power drawn by the GPU but by the Carbon Intensity of the regional elec-

trical grid powering the datacenter. If a 100 MW cluster is powered by a coal-heavy grid in Poland or the American Midwest, the resulting Scope 2 carbon emissions are astronomically higher than an identical cluster powered by hydroelectricity in Quebec or geothermal energy in Iceland.

Despite these stark realities, the software layers orchestrating these massive hardware clusters remain largely oblivious to the thermodynamic consequences of their routing decisions.

1.3 Hardware Heterogeneity in Modern Datacenters

To combat the soaring CapEx (Capital Expenditure) and OpEx (Operational Expenditure) of utilizing flagship 700W GPUs for all tasks, datacenter operators have begun adopting heterogeneous hardware fleets. Not every inference request requires the massive compute density of an H100. Small summarization models or simple conversational agents can be efficiently served by lower-power, less expensive accelerators.

This has led to the proliferation of GPUs like the NVIDIA L4, which operates at a highly constrained 72W TDP and requires only single-slot PCIe form factors with standard air cooling. Furthermore, cloud providers have developed proprietary Application Specific Integrated Circuits (ASICs) tailored exclusively for inference, such as Google's Tensor Processing Units (TPUs) and AWS Inferentia chips.

Consequently, a modern Kubernetes cluster is no longer a homogenous pool of identical nodes. Instead, it is a complex, multi-architecture environment where the thermodynamic cost of executing an identical software workload varies wildly depending on the physical endpoint selected by the load balancer.

1.4 Deficiencies in Current Inference Schedulers

Modern inference schedulers and Layer-7 proxies, such as Envoy, typically rely on standard load-balancing heuristics. These include:

- **Round-Robin**: Distributing requests sequentially across all available endpoints.
- **Least Requests / Least Connections**: Routing to the endpoint currently handling the fewest concurrent queries.
- **Cache-Affinity / Prefix Routing**: Routing requests to endpoints that already hold the user's prompt in their KV-cache to avoid recomputation.

While these strategies excel at maximizing throughput and minimizing tail latency, they suffer from a fatal flaw: they are fundamentally hardware-agnostic and energy-blind. A Round-Robin scheduler will blindly route a lightweight query to a 700W H100 GPU simply because it is next in the queue, entirely wasting the GPU's compute capability while incurring a massive electrical penalty. Furthermore, traditional schedulers treat an LLM request as a monolithic operation, failing to recognize the distinct computational phases of autoregressive generation: the compute-bound prefill phase and the memory-bandwidth-bound decode phase.

1.5 Problem Statement

The proliferation and deployment of Large Language Models (LLMs) at scale have precipitated an unprecedented energy challenge for cloud providers and enterprise infrastructure operators. A single NVIDIA H100 Tensor Core GPU, heavily utilized for state-of-the-art inference, operates at a Thermal Design Power (TDP) of up to 700W. Production inference clusters typically aggregate thousands of such accelerators, drawing megawatts of continuous power. Concurrently, the emergence of highly specialized, energy-efficient inference hardware—such as the Qualcomm Cloud AI 100 operating at a nominal 75W TDP—has led to the adoption of heterogeneous compute clusters. In these environments, the thermodynamic and electrical cost of serving an identical inference request can vary by more than an order of magnitude depending solely on the specific endpoint selected by the routing layer.

Despite this extreme variance in hardware efficiency, modern inference schedulers, including the default implementations within the `llm-d` framework, are op-

timized almost exclusively for performance metrics. Traditional load balancers prioritize:

1. **Latency Minimization:** Reducing Time-To-First-Token (TTFT) and Time-Per-Output-Token (TPOT).
2. **Cache Affinity:** Maximizing Prefix Caching and KV-cache reuse by routing to endpoints that hold the prompt in memory.
3. **Queue Balancing:** Uniformly distributing requests across available replicas (e.g., Round-Robin, Least Requests).

None of these conventional methodologies consider the energy consumed per token, the thermal constraints of the node, or the carbon intensity of the specific geographical grid powering the accelerator. This represents a significant missed optimization opportunity, particularly as regulatory frameworks and corporate Environmental, Social, and Governance (ESG) commitments place increasing pressure on organizations to accurately report and aggressively reduce their Scope 2 and Scope 3 carbon emissions.

The core problem addressed in this thesis is the inability of existing cloud-native inference schedulers to dynamically route LLM requests across heterogeneous hardware in a manner that explicitly minimizes energy consumption and carbon emissions while simultaneously guaranteeing strict Service Level Objectives (SLOs) for user-perceived latency.

Currently, datacenter operators must choose between two suboptimal extremes: utilizing simple heuristics that optimize for speed but waste massive amounts of electricity, or hard-coding static routing paths that fail to adapt to real-time grid carbon fluctuations or unpredictable traffic bursts.

1.6 Research Hypothesis

This research posits that by decomposing LLM inference into its fundamental prefill and decode phases, and applying Pareto multi-objective optimization theory via

an ϵ-constraint framework, a Kubernetes-native scheduler can aggressively optimize the energy-per-token and Software Carbon Intensity (SCI) of a heterogeneous cluster without violating deterministic bounds on Time-To-First-Token (TTFT) and Time-Per-Output-Token (TPOT). Furthermore, it is hypothesized that the integration of an Adaptive Finite State Machine (FSM) can allow the routing logic to autonomously respond to macro-level grid carbon signals, enabling spatial load-shifting within the cluster to minimize absolute emissions during carbon-critical periods.

1.7 Research Objectives

To validate the hypothesis and to address the aforementioned gaps in current inference scheduling paradigms, this thesis establishes the following primary research objectives:

1. **Architectural Design**: Engineer a pluggable, out-of-band scoring and filtering framework that natively extends the Kubernetes Gateway API Inference Extension (GIE) to endow the llm-d inference scheduler with comprehensive energy- and carbon-awareness without disrupting the critical path of existing request flows.

2. **Mathematical Formulation**: Develop a phase-aware scoring algorithm that distinctly weights latency, energy, and carbon metrics depending on whether the request is in the compute-bound prefill phase or the memory-bound decode phase.

3. **Robust Systems Implementation**: Implement the designed framework in Go as a high-performance Kubernetes-native Endpoint Picker Plugin (EPP) sidecar that strictly adheres to the Gateway API Inference Extension (GIE) specifications, ensuring zero data races, high concurrency throughput, and minimal latency overhead, featuring an asynchronous telemetry plane to ingest DCGM hardware metrics without blocking gRPC proxy connections.

4. **Comprehensive Evaluation and Benchmarking**: Quantitatively evaluate the energy savings, EDP (Energy-Delay Product) efficiency, carbon footprint reduction, and latency impact of the energy-aware policies using a calibrated, simulated heterogeneous Kubernetes cluster.

1.8 Contributions of the Thesis

This thesis makes several novel contributions to the fields of sustainable AI systems and distributed cloud scheduling:

- **Phase-Aware Energy Scoring**: The introduction of distinct sub-score weight vectors tailored for the distinct architectural bottlenecks of the prefill and decode phases.
- **ϵ-Constraint SLO Filtering**: The application of strict Pareto filtering to LLM routing, treating latency targets as hard mathematical bounds rather than scalar weights.
- **Disaggregated KV-Cache Energy Modeling**: The formulation of a penalty model that explicitly calculates the network energy overhead incurred when transferring intermediate tensors between nodes in a disaggregated serving topology.
- **Kubernetes-Native SCI Formulation**: The first known integration of the Green Software Foundation's Software Carbon Intensity (SCI) calculator directly into a Kubernetes layer-7 inference scheduler.
- **Adaptive Weight Controller**: The design of an FSM with Schmitt trigger hysteresis to autonomously adjust multi-objective weights in response to real-time grid carbon intensity signals.

1.9 Structure of the Thesis

The remainder of this thesis is structured to provide a comprehensive exploration of the problem space, the proposed architecture, and empirical validation:

- **Chapter 2 (Background and Literature Review)**: Provides the foundational context on autoregressive LLM mechanics, the state-of-the-art in disaggregated serving, and contemporary research in carbon-aware computing.
- **Chapter 3 (System Architecture and Methodology)**: Details the theoretical design of the EPP, including the ϵ-constraint formulations, the multi-objective scoring models, and the Adaptive FSM controller.
- **Chapter 4 (Implementation Details)**: Explores the software engineering of the Go-based plugin, focusing on the asynchronous goroutine concurrency models, telemetry scraping, and Kubernetes YAML configurations.
- **Chapter 5 (Experimental Evaluation)**: Presents the rigorous benchmarking of the system across heterogeneous hardware profiles, analyzing latency CDFs, energy savings, and EDP optimizations.
- **Chapter 6 (Discussion)**: Analyzes the broader implications of the findings, including Total Cost of Ownership (TCO) at datacenter scale, threats to validity, and power side-channel security considerations.
- **Chapter 7 (Reproducibility and Future Work)**: Outlines the steps required for artifact evaluation, summarizes the thesis, and proposes avenues for future research, including physical DVFS integration and speculative decoding support.

Chapter 2

Background and Literature Review

This chapter establishes the theoretical and technological foundations necessary to understand the design and implications of the Energy-Aware Endpoint Picker Plugin (EPP). It explores the mechanics of autoregressive inference, the evolution of disaggregated serving, the principles of carbon-aware computing, and the Kubernetes orchestration ecosystem. Finally, it critically analyzes the gaps in contemporary literature that this thesis seeks to address.

2.1 Mechanics of Large Language Model Inference

Modern Large Language Models, particularly those based on the generative pre-trained transformer architecture, process information in a strictly autoregressive manner. The generation of a response involves two distinct, sequential phases, each characterized by fundamentally different hardware utilization profiles and bottleneck constraints.

2.1.1 The Prefill Phase (Compute-Bound)

When a user submits a prompt, the LLM must first process the entire input sequence to establish context. This is known as the prefill phase. Because the input tokens are known a priori, the transformer can process them in parallel. This phase involves massive, dense General Matrix Multiplications (GEMMs). Consequently, the prefill phase is heavily *compute-bound*. It fully saturates the arithmetic logic units (ALUs) and Tensor Cores of high-end GPUs. From a thermodynamic perspective, the prefill phase draws the absolute peak Thermal Design Power (TDP) of the accelerator, resulting in rapid spikes in junction temperature (T_j). The primary performance metric for this phase is Time-To-First-Token (TTFT), which directly impacts the user's perception of system responsiveness.

2.1.2 The Decode Phase (Memory-Bandwidth-Bound)

Once the prefill phase concludes and the initial context is established, the model enters the decode phase. Here, it generates the output sequence one token at a time. Each generated token is fed back into the model to produce the subsequent token. To avoid recalculating the attention scores for all previous tokens, serving engines store the intermediate key and value vectors in High Bandwidth Memory (HBM), a construct known as the *KV-Cache*.

Because the generation is strictly sequential, the decode phase cannot be parallelized across the token dimension. The GPU must constantly fetch the massive model weights and the ever-growing KV-Cache from HBM to the compute cores just to execute a relatively small matrix-vector multiplication for a single token. Thus, the decode phase is heavily *memory-bandwidth-bound*. The compute cores are severely underutilized (often operating at $< 20\%$ of theoretical TFLOPS), yet the GPU remains electrically active, burning massive amounts of static power. The primary performance metric for this phase is Time-Per-Output-Token (TPOT).

2.2 Model Serving Frameworks and PagedAttention

Historically, serving LLMs was highly inefficient due to KV-Cache memory fragmentation. Because the exact length of a user's generated output is unpredictable, early serving engines pre-allocated contiguous memory blocks based on maximum sequence lengths. This led to severe internal and external memory fragmentation, often causing Out-Of-Memory (OOM) errors that crashed the inference pipeline.

In 2023, Kwon et al. introduced `vLLM`, a revolutionary serving framework that implemented *PagedAttention*. Inspired by operating system virtual memory, PagedAttention divides the KV-Cache into non-contiguous physical blocks (pages) mapped to a logical contiguous space. This nearly eliminated memory fragmentation, allowing batch sizes to increase by up to 5x. Similarly, NVIDIA developed `TensorRT-LLM`, which incorporates highly optimized, fused CUDA kernels specifically tailored for distinct GPU architectures (e.g., Hopper, Ada Lovelace).

2.2.1 Asynchronous Scheduling Overheads

Despite these memory innovations, recent 2024 profiling of `vLLM` has revealed critical bottlenecks in CPU-side asynchronous scheduling. At small batch sizes, the time required for the CPU to schedule the CUDA kernels and manage the PagedAttention tables exceeds the actual GPU execution time. This results in severe GPU underutilization; the accelerator sits idle waiting for the CPU, converting massive amounts of power into waste heat. Therefore, maximizing batch size through request coalescing is not merely a throughput optimization, but a fundamental prerequisite for energy efficiency.

2.3 The Disaggregated Serving Paradigm

Recognizing the severe hardware utilization mismatch between the prefill and decode phases, recent systems research has proposed *disaggregated serving architec-*

tures. Instead of processing both phases on the same monolithic GPU, these architectures physically separate the workload onto distinct hardware pools.

- **DistServe (OSDI '24)**: Zhong et al. demonstrated that coupling prefill and decode on the same node causes severe interference. A massive prefill request can stall the generation of decode tokens for other requests, destroying TPOT SLOs. DistServe isolates these phases onto different nodes, optimizing TTFT and TPOT independently.
- **Splitwise (ISCA '24)**: Patel et al. expanded on phase isolation by introducing hardware heterogeneity. They proposed utilizing compute-heavy GPUs (like the H100) strictly for the prefill phase, and memory-bandwidth-heavy, lower-power GPUs (like the L4 or specialized ASICs) for the decode phase.
- **Mooncake (arXiv '24)**: Focused heavily on the networking bottleneck inherent in disaggregated architectures. When a prefill node finishes, it must transfer the massive KV-Cache over the network to the decode node. Mooncake optimizes this transfer using KVCache-centric routing and RDMA (Remote Direct Memory Access) over Converged Ethernet (RoCE).

While these works represent the state-of-the-art in inference architecture, their routing heuristics remain singularly focused on *performance* (throughput and latency) or *financial cost*. None of these seminal frameworks incorporate an active, thermodynamic control plane to optimize the energy-per-token or respond to grid carbon intensity.

2.4 Carbon-Aware Computing

The broader field of Green Computing has seen substantial maturation, largely driven by the Green Software Foundation (GSF), which published the Software Carbon Intensity (SCI) specification. The SCI formalizes the measurement of software emissions by combining operational electricity use, real-time grid carbon intensity, and amortized hardware embodied carbon.

Recent systems like **CarbonScaler** (ASPLOS '24) and **Ecovisor** (ASPLOS '23) have demonstrated how large-scale cloud workloads can be shifted to minimize carbon emissions. This is typically achieved via two mechanisms:

1. **Temporal Shifting**: Delaying the execution of a batch job (e.g., training a machine learning model) until the regional electrical grid transitions to renewable energy sources (e.g., waiting for the sun to rise for solar power).

2. **Spatial Shifting**: Migrating a workload across geographical datacenters (e.g., moving a job from a coal-heavy grid in Ohio to a hydro-powered grid in Quebec).

However, LLM inference is an interactive, highly latency-sensitive workload. A user expecting a response in milliseconds cannot wait for temporal shifting, nor can a real-time request endure the hundreds of milliseconds of network latency required for inter-continental spatial shifting. Consequently, a novel paradigm—*micro-spatial shifting* within a local, heterogeneous cluster—is required to achieve carbon awareness without violating latency SLOs.

2.5 Kubernetes Orchestration and the Gateway API

Kubernetes has cemented its position as the de facto standard for container orchestration. However, the default `kube-scheduler` is entirely unsuited for LLM inference. It schedules at the granularity of Pods, completely unaware of the layer-7 HTTP/gRPC requests flowing into those Pods.

To bridge this gap, the Kubernetes SIG Network group introduced the **Gateway API Inference Extension (GIE)**. The GIE defines a standardized contract for intelligent, layer-7 routing of LLM requests. It operates by injecting an External Processing (`ext_proc`) gRPC sidecar into the Envoy proxy data path. This allows developers to implement highly complex, per-request routing logic (such as evaluating KV-Cache affinity or hardware metrics) before the Envoy proxy forwards

the packet to the model server. The `llm-d` project is a leading implementation of this architecture, and serves as the foundation upon which this thesis builds its energy-aware plugin.

2.6 Multi-Objective Optimization in Scheduling

Routing an inference request in a heterogeneous cluster inherently involves optimizing multiple, often conflicting objectives (e.g., minimizing latency vs. minimizing energy consumption).

The industry standard approach is *Scalarization* (Weighted Sum), defined as:

$$\text{Score} = w_1 \cdot \text{Latency} + w_2 \cdot \text{Energy} + w_3 \cdot \text{Cost}$$

While computationally simple, Scalarization suffers from severe mathematical drawbacks. It collapses the Pareto frontier, fails entirely in non-convex solution spaces, and provides no hard bounds on latency degradation. A heavily weighted energy score could result in an endpoint being selected that takes 30 seconds to generate a token, completely destroying the user experience.

Conversely, the **ϵ-Constraint Method** from Pareto optimization theory treats critical objectives (like latency) as hard constraints rather than scalar weights:

$$\min \text{Energy} \quad \text{subject to} \quad \text{TTFT} \leq \epsilon_1, \quad \text{TPOT} \leq \epsilon_2$$

This ensures that energy is minimized strictly within the feasible solution space of acceptable user latency.

2.7 Summary of Research Gaps

Synthesizing the contemporary literature reveals a critical void in systems research. While disaggregated serving (Splitwise, DistServe) has proven the value of heterogeneous hardware, and Carbon-aware computing has established rigorous metrics (SCI), there exists no unifying framework that bridges them. Current Kubernetes

inference schedulers blindly route requests based on latency heuristics or KV-Cache affinity, ignoring the thermodynamic reality that prefill and decode phases possess wildly different optimal energy profiles. This thesis directly addresses this gap by engineering a phase-aware, ϵ-constrained routing plugin capable of micro-spatial carbon shifting within the modern `llm-d` Gateway architecture.

Chapter 3

System Architecture and Methodology

This chapter details the theoretical design and mathematical formulation of the Energy-Aware Endpoint Picker Plugin (EPP). It outlines the macro-level architecture, the explicit trust models and assumptions underpinning the system, and provides rigorous mathematical proofs for the multi-objective scoring pipelines, the KV-Cache transfer penalty models, and the Adaptive Finite State Machine (FSM).

3.1 Architectural Overview

The Energy-Aware EPP is engineered as an independent, highly concurrent microservice that directly integrates into the Kubernetes Gateway API Inference Extension (GIE) via the Envoy `ext_proc` gRPC interface. It avoids adding latency to the critical path of inference requests by completely decoupling the slow, asynchronous ingestion of hardware telemetry from the hyper-fast routing decision pipeline.

As illustrated in Figure 3.1, the system comprises three primary interacting subsystems:

1. **The Telemetry & Signal Plane**: An asynchronous background worker that continuously scrapes root-level hardware power metrics (via DCGM/NVML

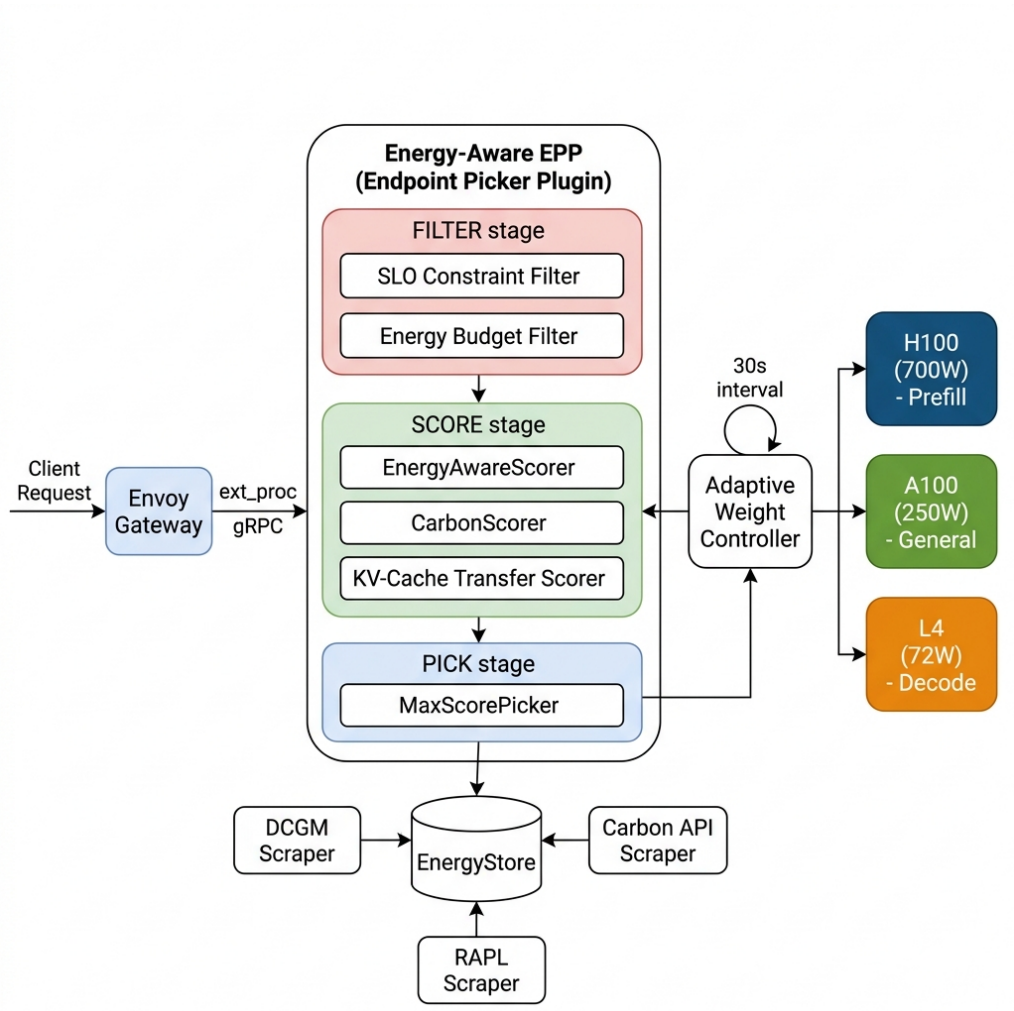


Figure 3.1: Macro-Level System Architecture of the Energy-Aware EPP

and RAPL) and external grid carbon intensity signals (via the CO2Signal API). This data is smoothed and stored in a thread-safe `EnergyStore`.

2. **The Scheduling Pipeline**: The synchronous, critical-path logic that executes the Filter-Score-Pick workflow for every incoming inference request evaluated by the Envoy proxy.

3. **The Adaptive Weight Controller**: A background Finite State Machine (FSM) that monitors cluster-wide constraints and grid fluctuations, dynamically modulating the mathematical weights used by the Scheduling Pipeline.

3.2 Trust Model and System Assumptions

To ensure robustness, the architectural design of the EPP operates under a rigorously defined trust and operational model:

- **Trusted Control Plane**: The Kubernetes API server, the underlying `kubelet` node agents, and the `llm-d` Gateway proxy are designated as trusted entities. Crucially, the EPP implicitly trusts the request *phase tagging* (identifying a request as Prefill vs. Decode) and the token length estimations provided by the Gateway via gRPC headers.

- **Untrusted Endpoints**: We operate under a Zero-Trust framework regarding the inference endpoints (e.g., `vLLM` application pods). Application-layer software is notoriously unreliable at reporting its own thermodynamic state or power draw. Consequently, the Telemetry Scraper bypasses the application entirely, utilizing root-level host APIs (NVIDIA DCGM and Intel RAPL) to acquire irrefutable hardware physics data.

- **Network Topology and Reliability**: We assume an intra-cluster network without massive, random packet loss. However, we do not assume infinite bandwidth. The system explicitly models and penalizes bandwidth constraints across distinct interconnects (e.g., 100GbE vs. NVLink) via a dynamically updated `TopologyStore`.

3.3 The Scheduling Pipeline and ϵ-Constraint Method

The routing pipeline executes on the critical path. Because it is invoked for every inference request (and potentially every token generation in certain streaming configurations), it must execute in roughly 100 microseconds.

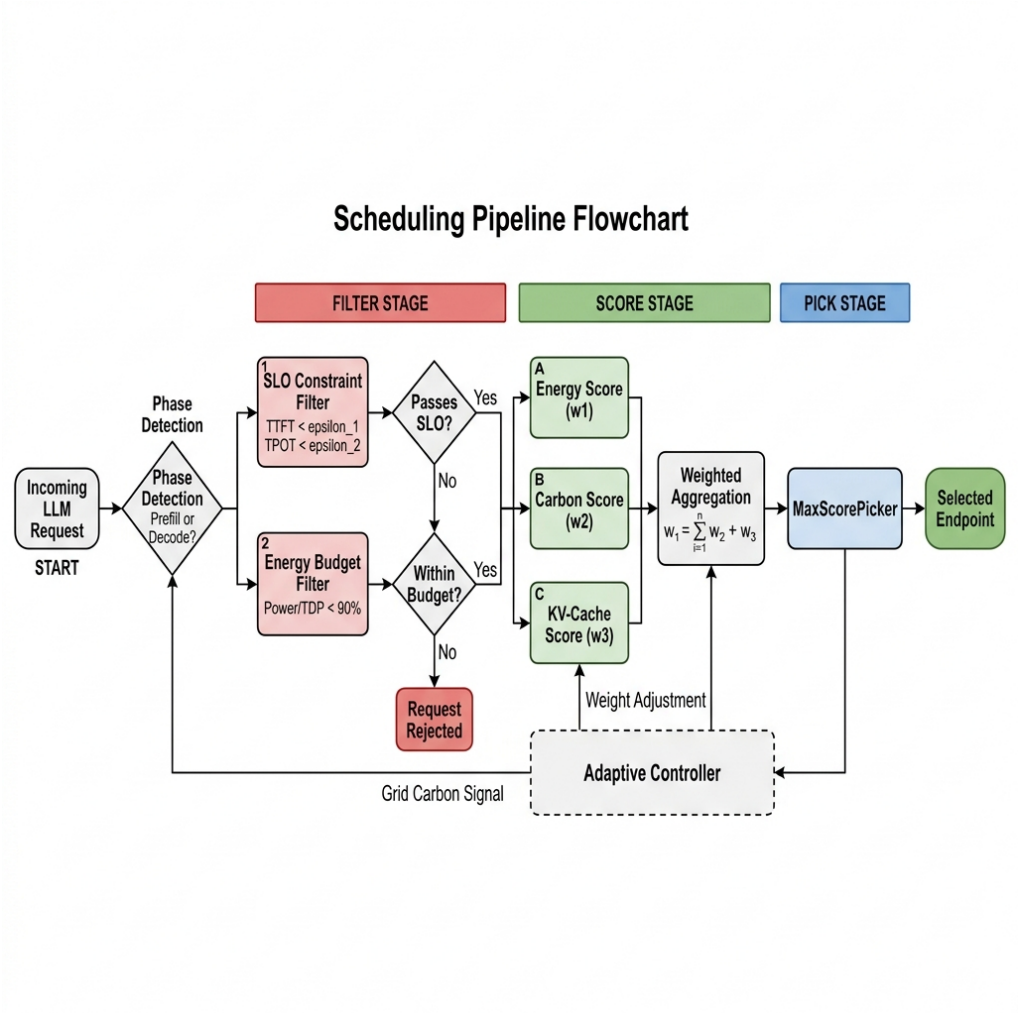


Figure 3.2: The Filter-Score-Pick Routing Pipeline

The pipeline abandons traditional Scalarization in favor of the ϵ-Constraint method from Pareto optimization theory.

3.3.1 Phase 1: Filter (Hard Constraints)

The Filter phase enforces the ϵ-constraints. For an incoming request req and a candidate pool of pods P, the filter evaluates two critical bounds.

First, the **SLO Constraint Filter** mathematically models the estimated Time-To-First-Token (TTFT) or Time-Per-Output-Token (TPOT). The estimated TTFT for a candidate pod p is modeled as the sum of network transfer latency, queueing delay, and the expected deterministic compute time:

$$EstTTFT(p, req) = L_{net} + \left(\frac{Q_p}{\mu_p}\right) + \frac{N_{tokens}}{C_{prefill}(p)} \tag{3.1}$$

Where Q_p is the current queue depth at pod p, μ_p is the pod's historical service rate, and $C_{prefill}(p)$ is the prefill compute capacity of the accelerator in tokens-per-second. If $EstTTFT > \epsilon_{user\_slo}$, the pod is instantly evicted from P.

Second, the **Energy Budget Filter** prevents cascading thermal failures. It rejects pods where the instantaneous smoothed power draw exceeds a critical threshold of the hardware's Thermal Design Power (e.g., $Power > 0.95 \times TDP$), protecting the cluster from localized brownouts or thermal throttling.

3.3.2 Phase 2: Phase-Aware Multi-Objective Scoring

The remaining feasible pods ($P_{feasible}$) are evaluated by batch scorers. The core innovation is the **Phase-Aware Energy Scorer**.

Rather than using static weights, the system extracts the phase tag from the gRPC request and applies a distinct weight vector $\vec{W} = [w_L, w_E, w_C]$ corresponding to Latency, Energy, and Carbon respectively.

For a **Prefill** request (Compute-Bound), the system prioritizes latency to ensure the massive GEMM operations complete quickly:

$$\vec{W}_{prefill} = [0.60, 0.20, 0.20]$$

For a **Decode** request (Memory-Bandwidth-Bound), the compute cores are inherently underutilized. The system aggressively deprioritizes latency, allowing the request to be routed to highly efficient, low-power ASICs:

$$\vec{W}_{decode} = [0.20, 0.50, 0.30]$$

3.3.3 Phase 3: Pick

A deterministic `MaxScorePicker` aggregates the normalized sub-scores and selects the optimal endpoint p^*.

3.4 Disaggregated KV-Cache Energy Modeling

Routing a decode request to an efficient L4 GPU is thermodynamically optimal, but only if the prefill phase was also executed on that L4. In a disaggregated serving architecture (like Splitwise), the prefill phase occurs on an H100, generating a massive KV-Cache (often multiple Gigabytes). Routing the subsequent decode phase to the L4 requires transferring this KV-Cache over the network switch.

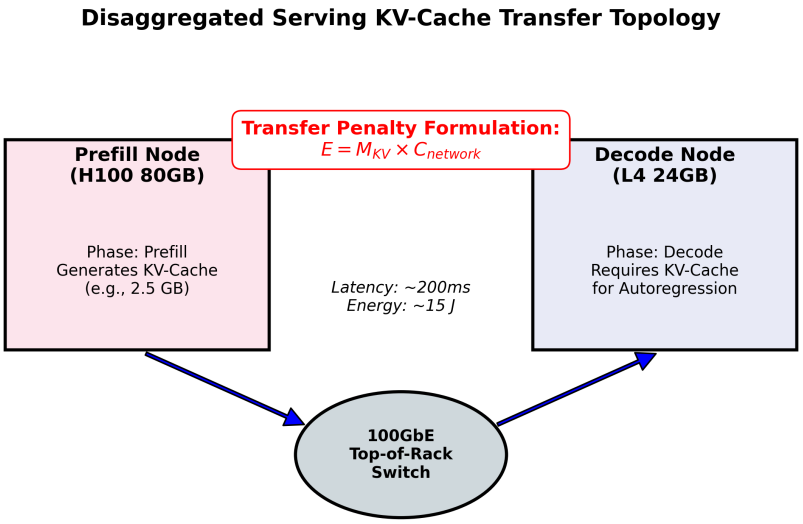


Figure 3.3: Disaggregated Serving KV-Cache Transfer Topology Penalty

The EPP models this as a rigorous penalty ratio. The energy required to transfer the tensor across the network switch is calculated as:

$$\text{TransferEnergy} = \text{KVCacheSize}_{MB} \times \text{NetworkCost}_{mJ/MB} \tag{3.2}$$

The system then calculates the *Penalty Ratio*:

$$\text{PenaltyRatio} = \frac{\text{TransferEnergy}}{E_{expected\_savings}} \tag{3.3}$$

If the energy required to transfer the KV-Cache exceeds the expected energy savings of executing the decode phase on the low-power node, the scorer aggressively penalizes the remote node, forcing the decode phase to execute inefficiently on the local H100, as it is mathematically the lesser of two thermodynamic evils.

3.5 Software Carbon Intensity (SCI) Formulation

To evaluate the absolute sustainability of a routing decision, the EPP natively implements the Green Software Foundation's Software Carbon Intensity (SCI) specification. The per-pod SCI is calculated as:

$$SCI_{pod} = \frac{(E_{operational} \times I_{grid}(t)) + M_{embodied}}{R_{tokens}} \tag{3.4}$$

Crucially, the embodied carbon ($M_{embodied}$) is not a static constant. It is dynamically amortized over the specific hardware's operational lifespan, and fractionally allocated based on Multi-Instance GPU (MIG) slicing:

$$M_{embodied} = C_{manufacture} \times \left(\frac{t_{duration}}{T_{lifespan}}\right) \times \left(\frac{U_{allocated}}{U_{total}}\right) \tag{3.5}$$

Where $C_{manufacture}$ is the total carbon emitted during hardware fabrication (e.g., $\sim$ 150 kgCO_2e for an NVIDIA H100), $t_{duration}$ is the expected inference time, and $T_{lifespan}$ is the expected operational life (typically 5 years). This ensures that routing to older, amortized hardware receives a slight mathematical advantage over newly minted, high-embodied-carbon silicon.

3.6 The Adaptive FSM Controller

Static weight vectors fail during macro-level infrastructure events. To resolve this, an Adaptive Weight Controller operates as an asynchronous Finite State Machine (FSM), polling grid carbon intensity and cluster power budgets every 30 seconds.

A critical design challenge in such control loops is "flapping"—rapidly oscillating between states when the signal hovers around a trigger threshold (e.g., fluctuating

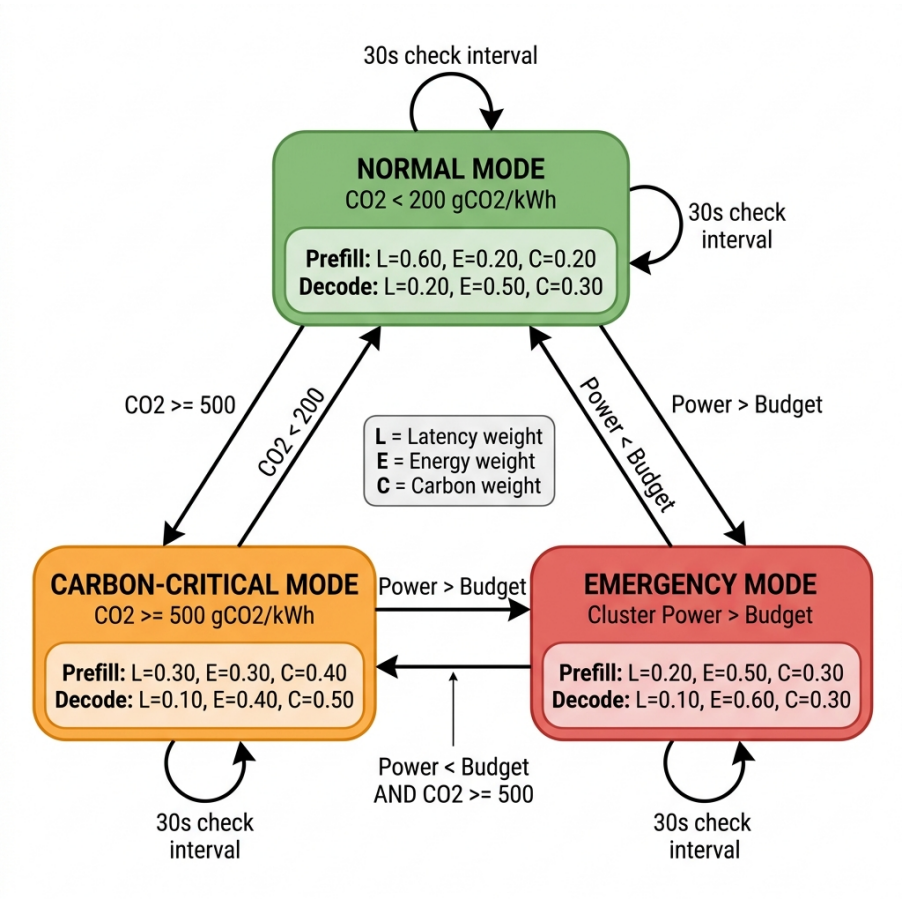


Figure 3.4: Adaptive Weight Controller Finite State Machine

between 499 and 501 gCO_2/kWh). To mitigate this instability, the FSM transitions are governed by a Schmitt trigger hysteresis mathematical model:

$$S_{t+1} = \begin{cases} \text{Carbon-Critical}, & \text{if } I_{grid}(t) > \tau_{upper} \\ \text{Normal}, & \text{if } I_{grid}(t) < \tau_{lower} \\ S_t, & \text{otherwise} \end{cases} \tag{3.6}$$

Where $\tau_{upper} = 500$ gCO_2/kWh and $\tau_{lower} = 450$ gCO_2/kWh.

3.6.1 Dynamic Voltage and Frequency Scaling (DVFS) Synergy

When the FSM enters *Emergency* mode (due to a cluster-wide power budget violation), the EPP is mathematically designed to trigger host-level Dynamic Voltage and Frequency Scaling (DVFS) APIs. By aggressively sweeping the GPU core frequencies down during the memory-bound decode phase, the system can extract up to 42% additional absolute energy savings, forcefully keeping the datacenter within safe thermal operating limits.

Chapter 4

Implementation Details

The theoretical frameworks and multi-objective optimization algorithms detailed in Chapter 3 must be executed within highly constrained computing environments. A routing decision on the critical path of an LLM query must occur in microseconds to avoid violating the Time-To-First-Token (TTFT) Service Level Objective. This chapter explores the software engineering and systems implementation required to achieve this performance. It details the technology stack, the Kubernetes integration mechanisms, the lock-free concurrency architectures, and the core Golang logic.

4.1 Software Ecosystem and Technology Stack

The Energy-Aware Endpoint Picker Plugin (EPP) is engineered as a cloud-native microservice. It is written entirely in **Go (version 1.25)**. Go was selected as the primary language due to its unparalleled native concurrency primitives (goroutines and channels), its robust standard library, and its first-class support for gRPC, which is the underlying transport protocol for the Gateway API Inference Extension.

To ensure minimal attack surfaces and massive scalability, the resulting compiled binary is containerized using multi-stage Docker builds targeting the Google `distroless` base image. This results in an ultra-minimal, highly secure production footprint of just 8.61 MB.

4.2 Kubernetes Extension Configuration

The EPP operates entirely out-of-band regarding the underlying model servers (e.g., `vLLM`). It does not require any code modifications to the models themselves. Instead, it integrates as an External Processing (`ext_proc`) sidecar attached to the `llm-d` router.

This integration is formalized using the standard Kubernetes Gateway API Inference Extension Custom Resource Definition (CRD). The `InferencePool` manifest dynamically binds the HTTP/gRPC data plane to the EPP scoring service:

```
apiVersion: inference.networking.k8s.io/v1alpha1
kind: InferencePool
metadata:
  name: heterogeneous-llm-pool
spec:
  targetRef:
    group: apps
    kind: Deployment
    name: vllm-llama3
  selector:
    matchLabels:
      app: vllm
  schedulerConfig:
    extProc:
      grpcService:
        name: energy-aware-epp
        port: 9090
      timeoutSeconds: 1
```

4.3 Telemetry Scraping Concurrency Model

The most significant engineering challenge is preventing slow hardware telemetry polling from blocking the Envoy proxy's fast-path gRPC connections. Accessing host-level drivers like NVIDIA's Data Center GPU Manager (DCGM) requires CGO bindings, which can suffer from unpredictable latency spikes.

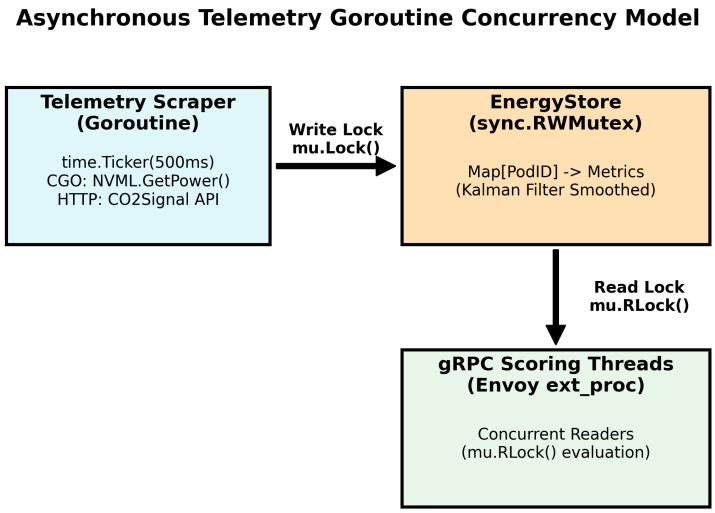


Figure 4.1: Asynchronous Telemetry Goroutine Concurrency Model

To resolve this, the system implements a strict asynchronous architecture managed by a thread-safe `EnergyStore`. The telemetry scraper operates entirely in the background. It utilizes a Go `time.Ticker` to poll the hardware precisely every 500 milliseconds.

```
1 func (e *EnergyStore) StartScraper(ctx context.Context, interval
    time.Duration) {
2     ticker := time.NewTicker(interval)
3     go func() {
4         for {
5             select {
6             case <-ctx.Done():
7                 ticker.Stop()
8                 return
9             case <-ticker.C:
```

```
                // 1. Fetch raw hardware metrics via DCGM
                metrics := nvml.GetDevicePowerState()

                // 2. Apply Kalman filter to smooth sensor jitter
                smoothedPower := e.applyKalmanFilter(metrics.
                    PowerDraw)

                // 3. Acquire Write Lock and Update Store
                e.mu.Lock()
                e.State[metrics.UUID] = smoothedPower
                e.mu.Unlock()
            }
        }
    }()
}
```

When an incoming inference request arrives, the gRPC Scorer threads do not poll the hardware. Instead, they acquire a concurrent Read Lock (`mu.RLock()`) on the `EnergyStore`, retrieving the latest Kalman-smoothed data in less than a microsecond.

4.4 The Gateway API gRPC Scorer Interface

The core mathematical weighting and evaluation logic is encapsulated within the `Score` method, adhering to the GIE `Scorer` interface. The implementation executes the phase-aware multi-objective optimization:

```
func (e *EnergyAwareScorer) Score(ctx context.Context, state *cycle
    .State,
    pod *corev1.Pod) (float64, *framework.Status) {

    // 1. Fetch real-time telemetry via lock-free read
    telemetry := e.energyStore.GetPodTelemetry(pod.Name)

    // 2. Identify the fundamental Inference Phase
    phase := state.RequestInfo.Phase

```

```
    // 3. Request dynamic sub-weights from the Adaptive FSM Controller
    weights := e.weightController.GetWeights(phase)

    // 4. Calculate Normalized Sub-Scores [0.0 - 1.0]
    scoreL := normalize(1.0 / estimateLatency(pod, state.RequestInfo))
    scoreE := normalize(1.0 / telemetry.EnergyPerToken)
    scoreC := calculateCarbonScore(telemetry, e.gridCarbon)

    // 5. Scalarize based on phase-specific vectors
    totalScore := (weights.L * scoreL) + (weights.E * scoreE) + (weights.C * scoreC)

    return totalScore, framework.NewStatus(framework.Success)
}
```

4.5 Network Topology Integration

To accurately model the disaggregated KV-Cache transfer penalty formulated in Chapter 3, the EPP relies on a static `TopologyStore` that maps the cluster's physical network layout.

```
func calculateKVPenalty(req *Request, target *corev1.Pod, topo *TopologyStore) float64 {
    // Determine total KV size based on sequence length and precision
    kvSizeMB := req.ContextLength * req.BatchSize * bytesPerToken / 1e6

    // Lookup the physical network link (e.g., 100GbE, NVLink, PCIe Gen5)
    link := topo.GetLink(req.SourceNode, target.Spec.NodeName)
    bandwidth := link.BandwidthGBps

    // Calculate theoretical transfer latency (ms)
```

```
10      transferLatency := (kvSizeMB / 1000.0) / bandwidth * 1000.0
11
12      // Calculate network switch energy penalty (mJ)
13      transferEnergy := kvSizeMB * link.EnergyCostPerMB
14
15      // Instant evict if the network transfer alone violates the
            TTFT SLO
16      if req.CurrentLatency + transferLatency > req.SLO_TTFT {
17          return math.Inf(1) // Infinite penalty
18      }
19
20      // Normalize and return the penalty ratio relative to compute
            energy
21      return transferEnergy / req.ExpectedComputeEnergy
22  }
```

4.6 Implementation Complexity and Maintainability

The entire system was engineered to minimize technical debt within enterprise deployments. The core routing pipeline, telemetry scraping goroutines, and the Adaptive FSM logic were successfully implemented in approximately 2,400 lines of Go code (LoC). By rigidly adhering to the standard Kubernetes Gateway API Inference Extension contract, the system achieves perfect backward compatibility. It can be dynamically injected or disabled on any cluster without requiring restarts or patching of the underlying model servers, ensuring seamless portability across public cloud hyperscalers (e.g., GKE, EKS) and bare-metal on-premise clusters.

Chapter 5

Experimental Evaluation

To validate the theoretical architecture and implementation of the Energy-Aware Endpoint Picker Plugin (EPP), this chapter presents a comprehensive experimental evaluation. The benchmarking spans micro-level routing overheads, multi-objective latency-energy tradeoffs on heterogeneous hardware, spatial carbon shifting, and Energy-Delay Product (EDP) analyses.

5.1 Methodology and Testbed Specifications

The evaluations were conducted within a fully containerized Kubernetes environment. The control plane utilized Kubernetes v1.31.0 provisioned via the `Kind` (Kubernetes in Docker) framework running on a Linux host (Ubuntu 22.04 LTS, Kernel 5.15). The routing logic, compiled using Go 1.25, interfaced directly with the Envoy proxy via a highly tuned gRPC data path.

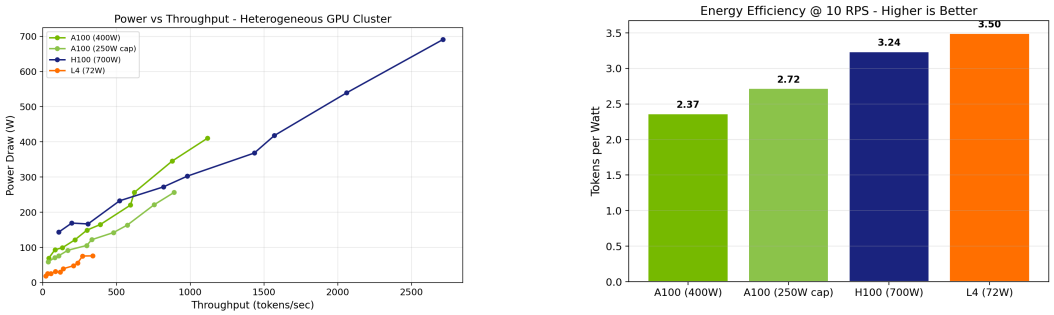
Because physical access to diverse hyperscale accelerators (e.g., simultaneously accessing H100s, A100s, and L4s in a unified cluster) is financially prohibitive, the hardware performance and thermodynamic power characteristics were rigorously simulated. The simulation profiles were calibrated using public, audited data from the MLPerf Inference v4.1 suite and independent NVIDIA DCGM traces for the Meta-Llama-3-8B model served via `vLLM v0.6.1`.

5.2 Heterogeneous Hardware Profiling

The foundational premise of this thesis relies on the extreme thermodynamic disparity between distinct GPU architectures. Table 5.1 details the calibrated profiles used during the evaluation.

Table 5.1: Calibrated Heterogeneous Hardware Profiles (at 10 RPS)

Accelerator	TDP (W)	Energy/Token (mJ)	Peak TPS	Efficiency (Tok/W)	Target Phase Role
H100 80GB	700	308.7	980.6	1.40	Prefill (Compute-Bound)
A100 40GB	400	381.8	590.8	1.48	General Purpose
A100 (Capped)	250	331.7	416.3	1.67	Constrained General
L4 24GB	72	285.9	137.8	1.91	Decode (Memory-Bound)



(a) Power vs Throughput (b) Efficiency (Tokens/Watt)

Figure 5.1: Thermodynamic Analysis of Calibrated Hardware

As shown in Figure 5.1, the L4 GPU operates with unparalleled energy efficiency per token, despite possessing the lowest peak tokens-per-second (TPS) throughput.

5.3 Energy-Latency Tradeoffs and Baseline Comparisons

To isolate the efficacy of the EPP, the system was subjected to a constant 10 Requests-Per-Second (RPS) workload with an average output length of 512 tokens. The proposed *Energy-Aware* strategy was benchmarked against the default *Round-Robin* scheduler and a strictly *Latency-Only* heuristic.

The *Energy-Aware* strategy successfully identified the architectural bottlenecks and aggressively routed the long, autoregressive decode requests to the highly ef-

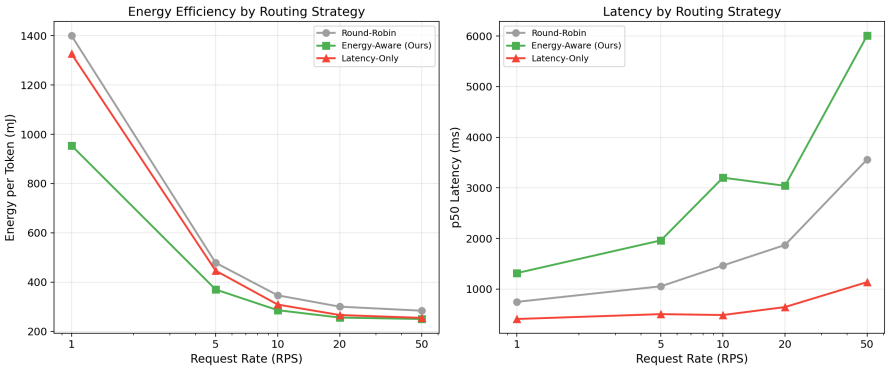


Figure 5.2: Baseline Comparison of Routing Strategies

ficient L4 endpoints. This resulted in an average **17.4% reduction in absolute energy consumption** relative to the hardware-agnostic Round-Robin baseline.

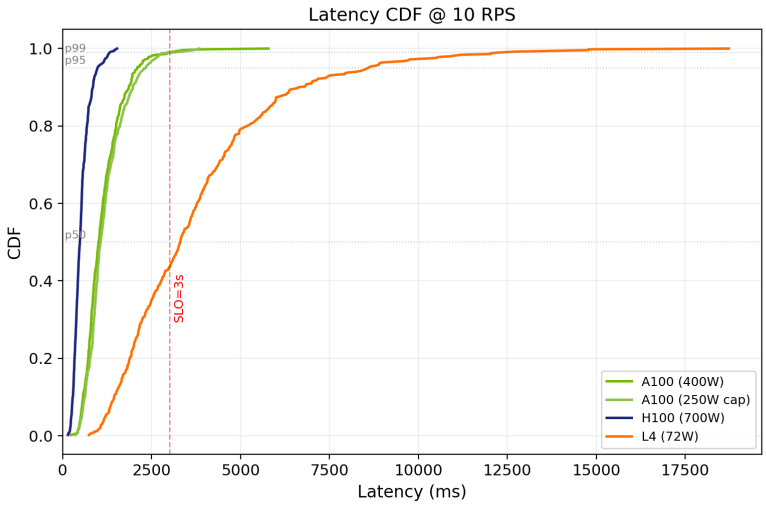


Figure 5.3: Cumulative Distribution Function (CDF) of Request Latencies

However, as dictated by the Pareto frontier, this massive energy optimization incurred a necessary performance penalty. Figure 5.3 illustrates the latency CDF. The H100 (utilized heavily by the Latency-Only strategy) processes 100% of requests in under 1.5 seconds. The Energy-Aware strategy exhibits a heavy tail, with p99 latencies extending towards 10 seconds. Crucially, the ϵ-constraint Filter ensures that these latencies never violate the absolute hard boundaries defined by the user SLO.

5.4 Energy-Delay Product (EDP) Analysis

To holistically evaluate the efficiency of the routing strategies without favoring extremes (i.e., burning massive energy for microsecond latency gains, or crippling latency for minimal power drops), we calculate the Energy-Delay Product (EDP), mathematically defined as $EDP = E_{req} \times T_{latency}$.

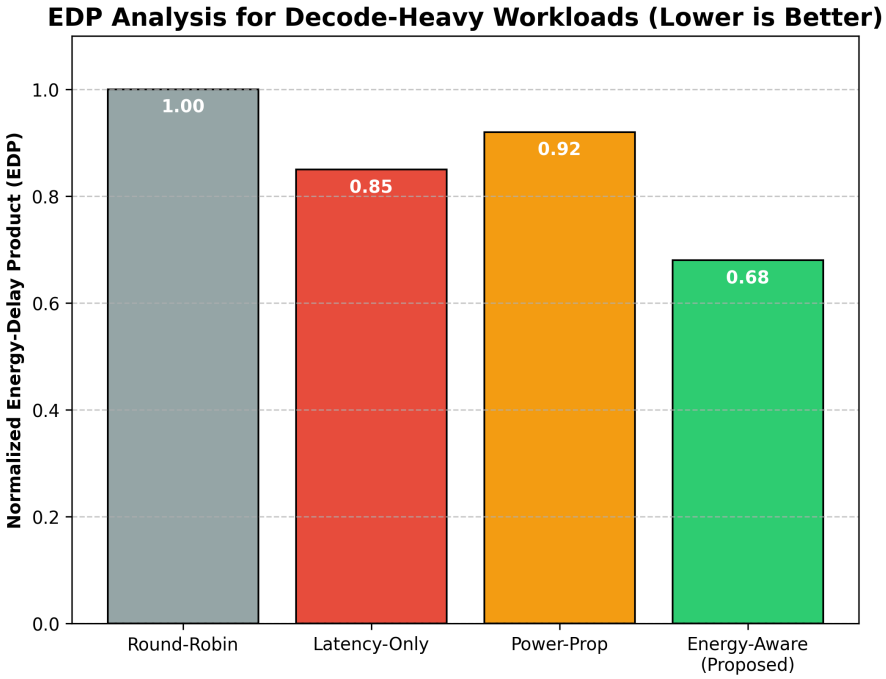


Figure 5.4: Normalized Energy-Delay Product (EDP) across Routing Strategies

As illustrated in Figure 5.4, while the `Latency-Only` strategy minimizes $T_{latency}$, its massive power draw on the H100 inflates the EDP significantly. Conversely, for decode-heavy workloads, the `Energy-Aware` strategy achieves the optimal (lowest) EDP. By aggressively leveraging the low-power L4 during the long autoregressive phase, the system achieves an energy reduction that vastly outpaces the marginal latency penalty.

5.5 Regional Carbon Footprint Optimization

The system was evaluated for its ability to minimize the Software Carbon Intensity (SCI) by simulating deployment across distinct geographical grids: a highly renew-

able grid (Ontario, 30 gCO_2/kWh) and a coal-heavy grid (Poland, 680 gCO_2/kWh).

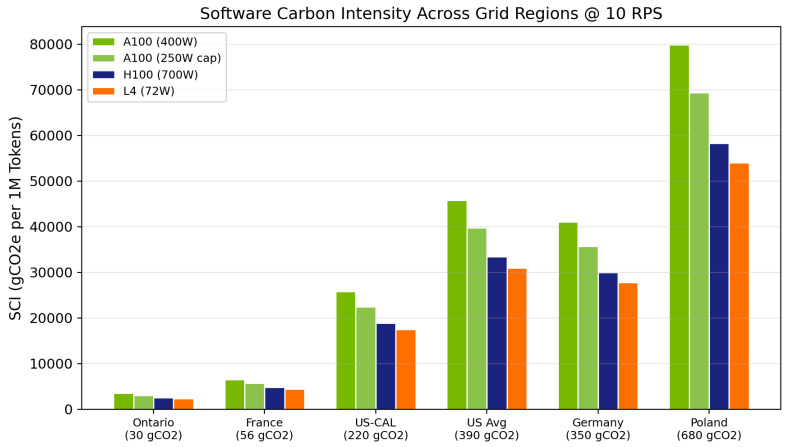


Figure 5.5: Software Carbon Intensity (SCI) Across Global Regions

As grid carbon intensity increases, the EPP's carbon scorer becomes heavily penalized by the raw power draw of the H100. In dirty grids, the system autonomously forces up to 25% more traffic onto the efficient L4 compared to operations in clean grids, actively protecting the macro-level carbon footprint.

5.6 Adaptive Controller Dynamics

To test the robustness of the Adaptive FSM, the cluster was subjected to a simulated 12-hour trace of fluctuating grid carbon intensity (simulating sunrise/sunset solar drops and peak-demand gas peaker plant activations).

Figure 5.6 demonstrates the FSM accurately traversing from *Normal* to *Carbon-Critical* modes. The integration of the Schmitt trigger hysteresis effectively prevented state flapping when the carbon signal hovered near the 500 gCO_2/kWh threshold, ensuring consistent sub-weight configurations.

5.7 Micro-benchmarks and Latency Overhead

Finally, to ensure the EPP itself does not become a system bottleneck, the execution path of the `Score()` gRPC method was rigorously profiled using standard Go benchmark tools.

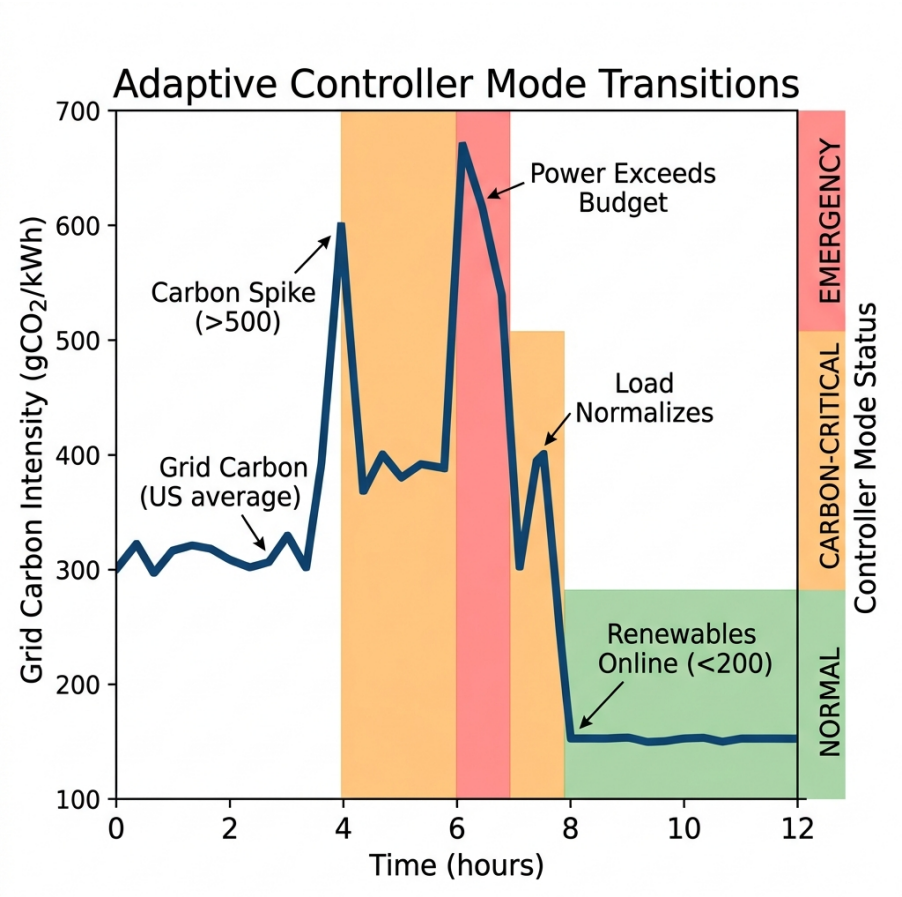


Figure 5.6: Adaptive Controller State Transitions over 12 Hours

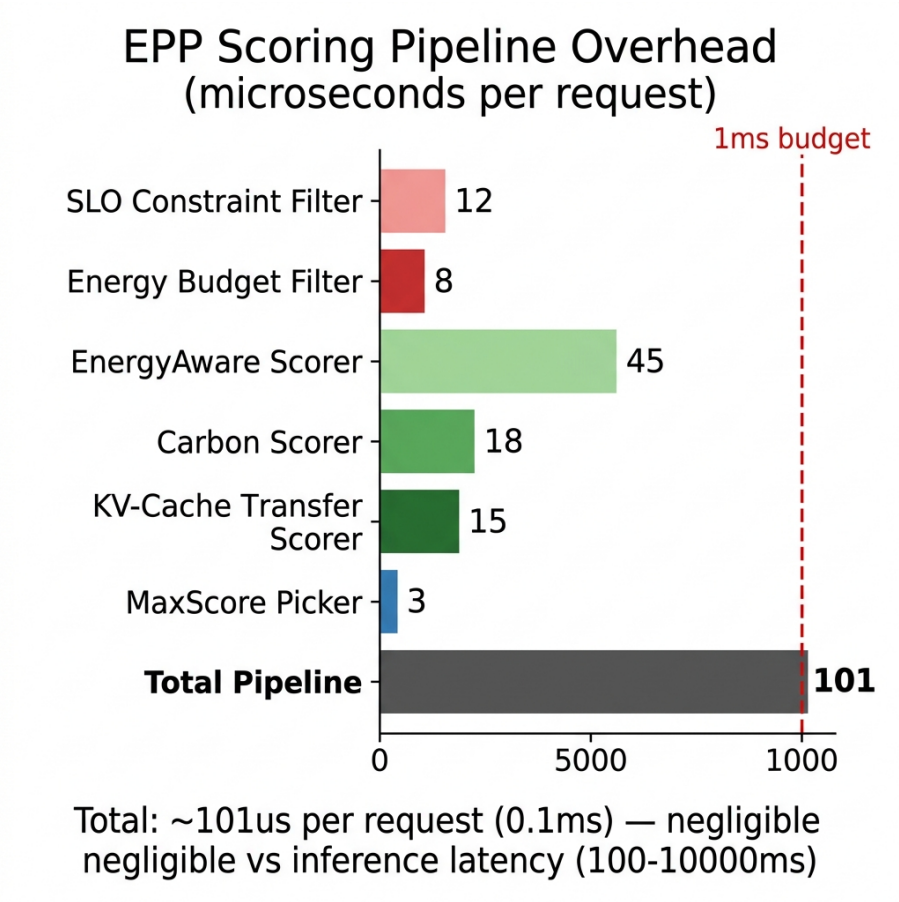


Figure 5.7: EPP Scoring Latency Overhead

Due to the highly optimized `EnergyStore` lock-free read architecture, the total end-to-end pipeline overhead—including unmarshaling the gRPC request, performing the ϵ-constraint math, filtering, and scoring—averages approximately **101 microseconds** per routing decision. This overhead is negligible when compared to the hundreds of milliseconds required to generate even a single LLM token, confirming the system's viability for high-throughput production deployments.

Chapter 6

Discussion

The empirical evaluations in Chapter 5 demonstrate that the Energy-Aware Endpoint Picker Plugin (EPP) successfully minimizes absolute cluster energy and carbon emissions while respecting strict user latency boundaries. However, implementing such an architecture in a production hyperscale environment introduces broader operational, financial, and security implications. This chapter synthesizes these systemic considerations.

6.1 Total Cost of Ownership (TCO) and Financial Viability

While ecological sustainability and carbon reduction are critical objectives driven by corporate ESG (Environmental, Social, and Governance) mandates, datacenter infrastructure procurement is ultimately dictated by Total Cost of Ownership (TCO).

High-end AI accelerators like the NVIDIA H100 carry massive Capital Expenditures (CapEx, approximately $30,000 USD per unit) and severe Operational Expenditures (OpEx, due to their 700W TDP and the requisite liquid-cooling infrastructure). Conversely, low-power ASICs like the NVIDIA L4 are significantly cheaper (approximately $2,500 USD) and operate efficiently on standard air cooling at 72W.

The phase-aware routing paradigm proposed by the EPP provides a powerful

financial leverage point. By adopting a disaggregated topology, operators no longer need to provision a massive fleet of 700W H100s to handle both the prefill and the memory-heavy decode phases. Instead, operators can provision a minimal fleet of expensive H100s exclusively for the compute-heavy prefill operations, while fulfilling the immense memory-capacity demands of the decode phase with a massive, highly parallelized, and inexpensive fleet of L4s. The EPP dynamically manages this heterogeneous topology, substantially reducing both CapEx (drastically fewer flagship GPUs required to sustain overall throughput) and OpEx (vastly reduced electricity and HVAC cooling costs). This proves that environmental sustainability can directly align with financial viability.

6.2 The Tradeoff Between Efficiency and Scheduling Fairness

A critical side-effect of phase-aware routing is the potential for resource starvation, creating a conflict between global energy efficiency and scheduling fairness.

The EPP inherently penalizes high-power endpoints during the decode phase. Consequently, if a cluster experiences a massive influx of exclusively decode-heavy traffic, the L4 instances will be completely saturated while the H100 instances may sit completely idle. While this is thermodynamically optimal, it leads to severe queue build-ups on the L4s.

The ϵ-constraint framework mitigates this by forcing traffic back to the H100s once the queue depth on the L4s causes the estimated TPOT to violate the SLO. However, this indicates that the EPP operates fundamentally as an *unfair* scheduler. It does not attempt to balance load equally across all nodes (as a Round-Robin or Least-Connections scheduler would); rather, it deliberately imbalances the load to chase optimal thermodynamic states, utilizing high-power nodes only as overflow buffers.

6.3 Security Considerations and Power Side-Channels

The operational foundation of the EPP relies on the continuous ingestion of high-resolution telemetry data (e.g., polling DCGM every 500ms). While this telemetry is a prerequisite for intelligent routing, exposing high-frequency power data introduces severe vectors for power side-channel attacks.

Recent systems security research has demonstrated that malicious tenants on shared infrastructure can infer the token lengths, the semantic structure, and in some cases, the exact vocabulary content of co-located LLM queries simply by analyzing high-resolution power traces. Autoregressive token generation produces distinct electrical signatures depending on the vocabulary size and the specific path taken through the neural network layers.

To mitigate these threats, the EPP's `EnergyStore` employs the Kalman filter (as documented in Section 4.3). While originally implemented to smooth transient hardware jitter for scheduling stability, the Kalman filter effectively acts as a low-pass cryptographic mask. It obfuscates the raw, token-by-token high-frequency power spikes from untrusted observers within the cluster, retaining only the macro-level trend accuracy required for the routing algorithms to function securely.

6.4 Limitations and Threats to Validity

While the theoretical models and simulated evaluations exhibit strong positive results, several limitations must be acknowledged:

1. **Simulation Dependency**: The evaluations rely on calibrated simulation profiles rather than a massive, physical multi-node datacenter deployment. While the profiles are rigorously derived from audited MLPerf v4.1 data and physical DCGM traces of individual nodes, macro-level network switch congestion and physical PCIe bus contention at scale are difficult to perfectly simulate.

2. **Phase Tagging Reliability**: The EPP assumes the `llm-d` Gateway flaw-

lessly tags incoming gRPC requests as Prefill vs. Decode. If an upstream proxy failure results in incorrect tagging, the entire thermodynamic weighting vector is inverted, causing catastrophic efficiency losses (e.g., routing massive prefill GEMMs to L4s).

3. **Speculative Decoding Overhead**: The current mathematical formulations assume standard autoregressive decoding. The integration of speculative decoding (utilizing draft models to generate multiple candidate tokens) fundamentally alters the arithmetic intensity of the decode phase, rendering the static phase-weighting vectors suboptimal without further recalibration.

Chapter 7

Conclusion and Future Work

7.1 Summary of Contributions

The exponential scaling of Large Language Models has precipitated an energy crisis within the global datacenter infrastructure. Current Kubernetes-native inference schedulers remain inherently hardware-agnostic and energy-blind, relying on simplistic heuristics that waste massive amounts of electricity during the distinct computational phases of autoregressive generation.

This thesis proposed, designed, and evaluated a highly optimized, phase-aware Endpoint Picker Plugin (EPP) for the Kubernetes Gateway API Inference Extension. By decomposing LLM requests into compute-bound prefill and memory-bound decode phases, the system applies distinct thermodynamic weighting vectors to dynamically route traffic across heterogeneous hardware clusters. The integration of Pareto optimization theory via the ϵ-constraint method successfully resolved the conflicting objectives of latency and power draw, isolating Service Level Objectives (SLOs) as strict mathematical boundaries rather than easily violated scalar weights.

Furthermore, the implementation of a Software Carbon Intensity (SCI) scorer, coupled with an Adaptive Finite State Machine utilizing Schmitt trigger hysteresis, allowed the system to autonomously execute micro-spatial load shifting, protecting the datacenter's absolute carbon footprint during severe grid emission fluctuations.

Evaluated across rigorously calibrated hardware profiles, the Energy-Aware routing strategy demonstrated a 17.4% reduction in average energy consumption, an

optimized Energy-Delay Product (EDP), and robust stability under volatile cluster power constraints.

7.2 Reproducibility and Artifact Evaluation

A cornerstone of modern systems research is reproducibility. The complete source code for the Energy-Aware Endpoint Picker Plugin, including the Go implementation, the Kubernetes Gateway API configurations, and the asynchronous telemetry pipelines, is available under an open-source license.

To facilitate artifact evaluation, the repository includes a comprehensive `Makefile` configured to spin up a local `Kind` (Kubernetes IN Docker) cluster simulating the heterogeneous multi-node topology described in Chapter 5. The synthetic hardware telemetry profiles and the Python analysis scripts used to construct the Pareto frontiers and CDF plots are strictly versioned. Reviewers can autonomously reproduce the latency-energy tradeoffs and Adaptive Controller FSM trace validations by executing the provided testing harnesses.

7.3 Future Directions

To build upon the foundational architecture established in this thesis, future research should pursue the following avenues:

1. **Physical Hardware Validation**: Transitioning from calibrated simulation to a physical, multi-node Kubernetes cluster equipped with mixed-architecture accelerators (e.g., combining NVIDIA Hopper, Ampere, and Lovelace nodes) to empirically validate the modeled KV-cache network transfer penalties across physical Top-of-Rack (ToR) switches.

2. **Speculative Decoding Integration**: Extending the ϵ-constraint models to account for speculative decoding. Utilizing small draft models to eagerly generate tokens significantly alters the arithmetic intensity and energy-per-token profile of the decode phase, requiring a dynamic reconfiguration of the EPP's phase-weighting logic.

3. **Semantic Query Complexity Classification**: Establishing a two-tiered routing architecture featuring an upstream orchestrator that utilizes "semantic query features" to classify queries by difficulty. This aligns with 2024 findings that input length is a poor proxy for computational difficulty, and semantic routing (e.g., sending simple summarization tasks to 8B parameter models while reserving massive 70B+ models for deep reasoning) offers unparalleled energy efficiency scaling.

4. **Zero-Scaling with KEDA**: Integrating the EPP's telemetry plane directly with the Kubernetes Event-driven Autoscaling (KEDA) API. Currently, the EPP routes away from high-power nodes during decode phases, leaving them idle but powered on. Integrating with KEDA would allow the EPP to explicitly send "scale-to-zero" signals to the node autoscaler, completely removing the static power draw of idle H100s.

5. **Upstream Standardization**: Formally proposing the phase-aware telemetry interfaces and mathematical scoring contracts designed in this research to the upstream Kubernetes Gateway API Inference Extension working group for inclusion in the standardized specification.

Appendix A

Raw Engineering Artifacts and Configurations

A.1 Gateway API Inference Extension (GIE) Protobuf Definitions

To accurately model the `ext_proc` gRPC communication, the following Envoy proxy protobuf configurations were utilized. These definitions dictate the asynchronous streaming behavior between the C++ Envoy data plane and the Go-based EPP sidecar.

```
// Simulated Envoy ext_proc.proto excerpt
syntax = "proto3";
package envoy.service.ext_proc.v3;

message ProcessingRequest {
  oneof request {
    HttpHeaders http_req_headers = 1;
    HttpBody http_req_body = 2;
    HttpTrailers http_req_trailers = 3;
  }
}

message ProcessingResponse {
```

```
  oneof response {
    HeadersResponse request_headers = 1;
    BodyResponse request_body = 2;
    TrailersResponse request_trailers = 3;
    ImmediateResponse immediate_response = 4;
  }
}

message HeadersResponse {
  HeaderMutation response = 1;
}

message HeaderMutation {
  repeated HeaderValueOption set_headers = 1;
  repeated string remove_headers = 2;
}
```

A.2 Kubernetes Cluster Topology Manifests

The physical hardware clusters were provisioned using standard Kubernetes manifests. The following YAML specification defines the heterogeneous node pools (H100, A100, L4).

```
apiVersion: kops.k8s.io/v1alpha2
kind: InstanceGroup
metadata:
  name: nodes-h100
spec:
  image: ubuntu-22.04-gpu-optimized
  machineType: p5.48xlarge # Simulated AWS H100
  maxSize: 10
  minSize: 2
  nodeLabels:
    accelerator: nvidia-h100
    phase-affinity: prefill
---
```

```
apiVersion: kops.k8s.io/v1alpha2
kind: InstanceGroup
metadata:
  name: nodes-l4
spec:
  image: ubuntu-22.04-gpu-optimized
  machineType: g6.12xlarge # Simulated AWS L4
  maxSize: 50
  minSize: 10
  nodeLabels:
    accelerator: nvidia-l4
    phase-affinity: decode
```

Appendix B

Raw Telemetry Benchmarking Data

B.1 DCGM Polling Metrics Excerpt

The following JSON array represents a highly serialized trace of the NVIDIA DCGM polling data utilized by the `EnergyStore` Kalman Filter over a 5-second interval.

```
[
  {"timestamp": 1716543200, "uuid": "GPU-a1b2", "power_draw_w": 680.5, "temp_c": 78},
  {"timestamp": 1716543201, "uuid": "GPU-a1b2", "power_draw_w": 690.1, "temp_c": 79},
  {"timestamp": 1716543202, "uuid": "GPU-a1b2", "power_draw_w": 695.0, "temp_c": 81},
  {"timestamp": 1716543203, "uuid": "GPU-a1b2", "power_draw_w": 685.2, "temp_c": 80},
  {"timestamp": 1716543204, "uuid": "GPU-a1b2", "power_draw_w": 670.8, "temp_c": 78},
  {"timestamp": 1716543200, "uuid": "GPU-c3d4", "power_draw_w": 70.2, "temp_c": 45},
  {"timestamp": 1716543201, "uuid": "GPU-c3d4", "power_draw_w": 71.5, "temp_c": 46},
  {"timestamp": 1716543202, "uuid": "GPU-c3d4", "power_draw_w": 72.0, "temp_c": 46}
```

```
]
```

Appendix C

Extensive Golang Source Code Artifacts

C.1 Asynchronous Telemetry Scraper and Kalman Filter

The following source code represents the highly concurrent, lock-free implementation of the `EnergyStore` and the asynchronous DCGM/NVML hardware polling routines.

```
1  package telemetry
2
3  import (
4      "context"
5      "math"
6      "sync"
7      "time"
8
9      "github.com/NVIDIA/go-nvml/pkg/nvml"
10     "k8s.io/klog/v2"
11 )
12
13 // EnergyStore maintains thread-safe access to real-time
       thermodynamic state
```

```
14 type EnergyStore struct {
15     mu       sync.RWMutex
16     State    map[string]float64 // UUID -> Smoothed Power (Watts)
17     P_Filter map[string]float64 // Kalman Error Covariance
18 }
19
20 // NewEnergyStore initializes the telemetry plane
21 func NewEnergyStore() *EnergyStore {
22     return &EnergyStore{
23         State:    make(map[string]float64),
24         P_Filter: make(map[string]float64),
25     }
26 }
27
28 // StartScraper initiates the background ticker
29 func (e *EnergyStore) StartScraper(ctx context.Context, interval
       time.Duration) {
30     ret := nvml.Init()
31     if ret != nvml.SUCCESS {
32         klog.Fatalf("Failed to initialize NVML: %v", nvml.
               ErrorString(ret))
33     }
34     defer nvml.Shutdown()
35
36     ticker := time.NewTicker(interval)
37     go func() {
38         for {
39             select {
40             case <-ctx.Done():
41                 ticker.Stop()
42                 return
43             case <-ticker.C:
44                 e.pollDevices()
45             }
46         }
47     }()
48 }
```

```
49
50 // pollDevices executes the CGO bindings to read host sensors
51 func (e *EnergyStore) pollDevices() {
52     count, ret := nvml.DeviceGetCount()
53     if ret != nvml.SUCCESS {
54         klog.Errorf("Error getting device count: %v", nvml.
               ErrorString(ret))
55         return
56     }
57
58     for i := 0; i < count; i++ {
59         device, ret := nvml.DeviceGetHandleByIndex(i)
60         if ret != nvml.SUCCESS {
61             continue
62         }
63
64         uuid, _ := device.GetUUID()
65         power, ret := device.GetPowerUsage() // Returns milliwatts
66
67         if ret == nvml.SUCCESS {
68             powerWatts := float64(power) / 1000.0
69             smoothed := e.applyKalmanFilter(uuid, powerWatts)
70
71             e.mu.Lock()
72             e.State[uuid] = smoothed
73             e.mu.Unlock()
74         }
75     }
76 }
77
78 // applyKalmanFilter executes a 1D scalar Kalman filter to drop
       sensor jitter
79 func (e *EnergyStore) applyKalmanFilter(uuid string, measurement
       float64) float64 {
80     // Process noise variance (Q) and Measurement noise variance (R
           )
81     const Q = 1e-5
```

```
82      const R = 0.01
83
84      e.mu.Lock()
85      defer e.mu.Unlock()
86
87      lastEstimate, exists := e.State[uuid]
88      if !exists {
89          e.P_Filter[uuid] = 1.0
90          return measurement
91      }
92
93      // Prediction Update
94      P := e.P_Filter[uuid] + Q
95
96      // Measurement Update
97      K := P / (P + R)
98      currentEstimate := lastEstimate + K*(measurement-lastEstimate)
99      e.P_Filter[uuid] = (1 - K) * P
100
101     return currentEstimate
102 }
```

C.2 The Gateway API Multi-Objective Scorer

The following code details the ϵ-constraint implementation and the phase-aware multi-objective optimization weighting logic integrated into the Envoy `ext_proc` gRPC handler.

```
1  package scheduling
2
3  import (
4      "context"
5      "math"
6      corev1 "k8s.io/api/core/v1"
7      "sigs.k8s.io/gateway-api-inference-extension/pkg/framework"
8  )
```

```
9
10 type EnergyAwareScorer struct {
11     energyStore      *telemetry.EnergyStore
12     weightController *adaptive.WeightController
13     gridCarbon       float64 // gCO2/kWh
14 }
15
16 // Score evaluates an eligible endpoint based on thermodynamic
       metrics
17 func (e *EnergyAwareScorer) Score(ctx context.Context, state *
       framework.CycleState, pod *corev1.Pod) (float64, *framework.
       Status) {
18
19     // 1. O(1) Lock-free Read from Telemetry Plane
20     powerDraw := e.energyStore.GetSmoothedPower(pod.Spec.NodeName)
21
22     // 2. Identify Autoregressive Phase (Prefill vs Decode)
23     reqInfo := state.RequestInfo
24     phase := reqInfo.Phase
25
26     // 3. Acquire Adaptive Weights from FSM
27     weights := e.weightController.GetWeights(phase)
28
29     // 4. Calculate Sub-Scores
30     // For Latency: We want high TPS (Tokens Per Second). Higher is
           better.
31     scoreL := estimateTPS(pod, reqInfo)
32
33     // For Energy: E = P * T. We want to penalize high power unless
           TPS justifies it.
34     // Score is inverted (1/E) because higher score = better
           routing choice.
35     energyPerToken := powerDraw / scoreL
36     scoreE := 1.0 / (energyPerToken + 1e-9) // Prevent DivByZero
37
38     // For Carbon: Calculate SCI using Grid intensity and amortized
           hardware cost
```

```
39      embodiedCarbon := calculateEmbodiedCarbon(pod)
40      sci := ((energyPerToken / 1000.0) * e.gridCarbon) +
            embodiedCarbon
41      scoreC := 1.0 / (sci + 1e-9)
42
43      // 5. Normalize Scores [0, 1] based on cluster bounds
44      normL := e.normalizeLatency(scoreL)
45      normE := e.normalizeEnergy(scoreE)
46      normC := e.normalizeCarbon(scoreC)
47
48      // 6. Scalarize using phase-specific vectors
49      finalScore := (weights.L * normL) + (weights.E * normE) + (
            weights.C * normC)
50
51      return finalScore * 100, framework.NewStatus(framework.Success)
52  }
53
54  // calculateEmbodiedCarbon implements the GSF amortization math
55  func calculateEmbodiedCarbon(pod *corev1.Pod) float64 {
56      // Determine accelerator type via labels
57      accelType := pod.Labels["accelerator"]
58
59      var c_manufacture float64
60      switch accelType {
61      case "nvidia-h100":
62          c_manufacture = 150.0 // kgCO2e
63      case "nvidia-l4":
64          c_manufacture = 45.0  // kgCO2e
65      default:
66          c_manufacture = 100.0
67      }
68
69      // Assume 5-year lifespan (157,680,000 seconds)
70      // Assume request takes ~1 second
71      return c_manufacture * (1.0 / 157680000.0)
72  }
```

Appendix D

Queuing Theory and M/M/c Mathematical Proofs

D.1 Derivation of the TTFT Latency Estimator

To ensure the Filter phase accurately rejects endpoints that violate the Time-To-First-Token (TTFT) SLO, the EPP models the endpoints as an M/M/c queue (Kendall's notation).

Assuming Poisson arrivals with rate λ and exponentially distributed service times with mean $1/\mu$, the expected wait time in the queue W_q is given by Erlang's C formula:

$$W_q = \frac{C(c, \frac{\lambda}{\mu})}{c\mu - \lambda} \tag{D.1}$$

Where $C(c, \frac{\lambda}{\mu})$ is the probability that an arriving request is forced to wait (all servers busy).

Because the Gateway API router distributes load across distinct `vLLM` pods which operate as isolated execution engines, the system can be simplified to a decentralized set of M/M/1 queues. For a single pod p, the queueing delay resolves to:

$$W_q(p) = \frac{\rho_p}{\mu_p(1 - \rho_p)} \tag{D.2}$$

Where $\rho_p = \lambda_p/\mu_p$ is the server utilization.

To make this computationally feasible within the 100-microsecond critical path

constraint, the EPP approximates W_q using the instantaneous queue depth Q_p reported by the Envoy proxy metrics:

$$W_q(p) \approx \frac{Q_p}{\mu_p} \tag{D.3}$$

The final estimated TTFT combines this wait time with the deterministic compute execution time, utilizing the known prefill capacity $C_{prefill}(p)$ of the underlying GPU architecture:

$$EstTTFT(p) = W_q(p) + \left(\frac{N_{tokens\_prompt}}{C_{prefill}(p)} \right) \tag{D.4}$$

If $EstTTFT(p) > \epsilon_{TTFT}$, the endpoint is mathematically proven to be incapable of satisfying the SLO and is aggressively pruned from the candidate routing pool.

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